

ZAMBOANGA, PhilippinesWMO: **988360**Lat: **6.922N**Lon: **122.060E**Elev: **10**StdP: **101.20**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	23.2	19.8	14.5	25.4	20.7	15.4	25.7	5.6	30.5	5.0	30.7	0.4	40	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.5	34.2	27.4	33.8	27.3	33.2	27.1	28.3	32.9	28.0	32.5	27.7	32.1	2.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	31.7	26.8	22.4	31.3	26.3	21.8	30.8	91.5	32.7	89.6	32.5	88.2	32.5	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.4	4.7	4.1	20.3	35.5	1.7	0.7	19.0	35.9	18.0	36.3	17.0	36.7	15.8	37.2		
(5)			19.3	29.5	1.2	0.6	18.4	29.9	17.7	30.3	17.1	30.6	16.2	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	27.9	28.1	28.4	28.7	28.8	28.4	28.1	28.3	28.2	28.1	28.3	28.3
	DBStd	0.92	0.89	0.82	0.82	0.78	0.91	0.99	0.97	0.93	0.94	0.92	0.84	0.79
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6684	556	507	571	561	582	552	562	567	547	562	550	567
	CDD18.3	3642	297	274	313	311	324	302	304	309	297	304	300	308
	CDH23.3	39556	3077	2919	3493	3563	3697	3294	3234	3363	3171	3231	3222	3291
	CDH26.7	15324	1157	1158	1447	1477	1504	1231	1167	1264	1186	1201	1254	1279
Wind	WSAvg	1.5	1.4	1.5	1.5	1.6	1.4	1.4	1.6	1.6	1.7	1.7	1.4	1.4
Precipitation	PrecAvg	1214	53	53	56	64	92	135	146	129	151	198	99	69
	PrecMax	1577	163	209	175	186	267	273	359	319	355	630	306	288
	PrecMin	769	0	0	0	0	3	16	15	18	9	20	13	5
	PrecStd	222	46	49	47	50	51	59	69	71	89	117	61	62
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	34.0	34.3	34.7	34.9	34.2	33.9	33.9	34.0	34.0	34.1	34.1
		MCWB	26.7	26.7	26.8	27.5	28.1	27.8	27.5	27.4	27.3	27.5	27.5	27.3
	2%	DB	32.9	33.1	33.6	33.8	33.9	33.2	32.9	33.0	33.0	33.1	33.2	33.1
		MCWB	26.7	26.5	26.7	27.2	27.7	27.4	27.3	27.3	27.2	27.3	27.3	27.0
	5%	DB	32.1	32.3	32.9	33.0	33.0	32.4	32.1	32.2	32.2	32.2	32.7	32.4
		MCWB	26.5	26.4	26.6	27.1	27.4	27.1	27.0	27.0	27.0	27.0	27.1	26.8
	10%	DB	31.5	31.8	32.1	32.2	32.1	31.7	31.4	31.8	31.6	31.6	31.9	31.8
		MCWB	26.3	26.3	26.5	26.9	27.1	27.0	26.8	26.9	26.8	26.8	26.8	26.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	27.7	28.1	28.2	28.8	28.6	28.3	28.2	28.2	28.2	28.5	28.1
		MCDB	32.3	32.5	33.0	33.0	33.6	33.3	32.6	32.6	32.4	32.6	33.0	32.8
	2%	WB	27.2	27.2	27.5	27.7	28.1	27.9	27.7	27.7	27.7	27.7	27.7	27.6
		MCDB	31.8	31.9	32.4	32.3	32.7	32.2	31.8	32.0	31.9	31.9	32.2	32.1
	5%	WB	26.9	26.9	27.1	27.5	27.7	27.5	27.2	27.3	27.3	27.3	27.4	27.2
		MCDB	31.3	31.5	31.9	32.0	32.0	31.6	31.3	31.6	31.4	31.4	31.7	31.6
	10%	WB	26.5	26.5	26.7	27.1	27.3	27.1	26.9	27.0	26.9	26.9	27.0	26.8
		MCDB	30.7	30.9	31.3	31.5	31.4	31.0	30.8	31.1	30.9	31.0	31.2	31.0
Mean Daily Temperature Range		MDBR	7.7	8.0	8.2	7.9	7.5	7.0	6.9	7.1	7.3	7.4	7.8	7.8
	5% DB	MCDBR	8.6	8.9	9.1	8.8	8.5	8.1	7.9	8.0	8.1	8.4	8.9	8.6
		MCWBR	4.1	4.3	4.3	4.1	4.0	3.9	3.8	3.8	3.9	4.1	4.2	4.1
	5% WB	MCDBR	8.0	8.2	8.4	8.2	8.1	7.6	7.4	7.7	7.7	7.9	8.3	8.1
		MCWBR	4.1	4.3	4.2	4.1	4.1	4.0	3.8	3.9	4.0	4.1	4.3	4.1
Clear-Sky Solar Irradiance	taub	0.419	0.425	0.427	0.439	0.411	0.402	0.404	0.432	0.430	0.442	0.417	0.418	
	taud	2.503	2.465	2.466	2.427	2.521	2.562	2.553	2.447	2.437	2.408	2.492	2.495	
	Ebn at Noon	898	900	897	872	879	876	877	867	881	873	893	892	
	Edh at Noon	109	116	117	119	106	100	102	115	118	121	109	108	
All-Sky Solar Radiation	RadAvg	5.05	5.62	5.96	5.92	5.35	4.91	4.91	5.15	5.15	5.08	5.13	4.97	
	RadStd	0.40	0.44	0.43	0.33	0.34	0.23	0.33	0.29	0.39	0.36	0.37	0.36	